Charger Test Methodology and Results

Test Plan for Charger:

Table 1: Test Plan and expected result

Test	Test Method	Expected Results
Test Points	Measure Vout, Vref and breaker circuitry before and after applying a voltage above the trip threshold	PCB measurements should be close to simulated and theoretical values, with Vout regulating at approximately 3.5 V up until the current limit before dropping off.
MOSFET Temperature	Measure voltage drop and current through mosfet to calculate power dissipated.	MOSFET temperature should remain within its safe operating temperature range.
Vout Voltage Ripple	Measure Vout using the oscilloscope and record the peak to peak voltage ripple.	Vout should remain stable around 3.5 V with less than 5% deviation (175mV).
Vref Protection	Increase the supply to the Vref circuit while monitoring Vref to test the Zener protection.	The Zener should clamp Vref and prevent it from increasing significantly above its intended value.
Current Limit	Continue decreasing load resistance until the current limit activates .	Output current should be limited to approximately 0.5 A at 7 ohms rather than continuing to increase.
Overvoltage Comparator	Slowly increase the voltage applied to the Vout sensing input while monitoring the comparator output. Record the voltage at which the comparator changes state.	The comparator output should switch towards the 5 V rail when Vout reaches approximately 3.7 V.
Sr Latch Set	With the latch in its reset state and the breaker off, press the push button connected to the Set input.	The SR latch should return to Q = 1, turning the breaker back on and reconnecting the circuit.

Output Values:

When testing the charger, we decided to conduct testing over a wide range of data points to give us an indication of whether our system was functional in many scenarios rather than just the ones we hand picked. To accommodate for unexpected changes in our input voltage we decided to test the circuit under 3 input conditions to see how it behaved.

Table 2: Output voltages and current at a 5V input

5V input		
Resistance	**Voltage**	**Amps**
100.000	3.495	0.034
90.000	3.495	0.038
80.000	3.493	0.043
70.000	3.493	0.049
60.000	3.492	0.057
50.000	3.490	0.069
40.000	3.487	0.086
30.000	3.483	0.114
20.000	3.474	0.170
10.000	3.430	0.327
9.000	3.426	0.361
8.000	3.418	0.403
7.000	3.412	0.458
6.000	3.152	0.489
5.000	2.660	0.488
4.000	2.180	0.489
3.000	1.692	0.490
2.000	1.200	0.490
1.000	0.730	0.490
0.500	0.375	0.494

We started off with the standard 5V input at a wide range of load resistances across the output. We can see output voltage remains relatively close to the 3.5V up until 20 ohms where we see a small drop in voltage. We can see that there is a sudden drop in voltage around 7 ohms which is as expected as this was our target current limiting resistance

We can see that the current starts to reach its limit around 6 ohms which also never exceeds the 0.5A current limit we set even while testing with a load resistance of 0.5 ohms. Our simulation and theoretical calculations provided us with a goal to test against. We expected the voltage to drop by 0.5V per ohm reduced on the load below 7 ohms due to the 0.5A limit on our output current. As shown in the table above our charger achieved this 0.5V drop but with slightly higher output voltage than we were expecting. The pattern seems to show that the circuit is receiving between 0.45-0.5 ohms of additional resistance added to the load resistance which is a pattern recognisable between 1-50 ohms at the output load.

Table 3: Output voltages and current at a 4.7V input

4.7V input		
Resistance	**Voltage**	**Amps**
100.000	3.413	0.034
90.000	3.412	0.037
80.000	3.413	0.042
70.000	3.412	0.048
60.000	3.412	0.056
50.000	3.410	0.067
40.000	3.410	0.084
30.000	3.400	0.112
20.000	3.390	0.167
10.000	3.360	0.329
9.000	3.351	0.363
8.000	3.340	0.407
7.000	3.330	0.462
6.000	2.940	0.473
5.000	2.470	0.474
4.000	2.000	0.474
3.000	1.530	0.475
2.000	1.055	0.476
1.000	0.580	0.476
0.500	0.340	0.476

Table 4: Output voltages and current at a 5.3V input

5.3V input		
Resistance	**Voltage**	**Amps**
100.000	3.546	0.035
90.000	3.545	0.039
80.000	3.545	0.044
70.000	3.546	0.050
60.000	3.545	0.058
50.000	3.543	0.070
40.000	3.541	0.088
30.000	3.536	0.117
20.000	3.528	0.174
10.000	3.500	0.342
9.000	3.493	0.379
8.000	3.486	0.424
7.000	3.477	0.481
6.000	3.120	0.501
5.000	2.620	0.502
4.000	2.120	0.502
3.000	1.620	0.502
2.000	1.110	0.502
1.000	0.610	0.502
0.500	0.360	0.503

The linear regulator's job is to regulate a constant output voltage even with slight changes in the input voltage. We repeated our measurements both at an input voltage of 4.7V and 5.3V to see how the charger responded. We can see that the current limiting set in around 7 ohms for both input voltages but had an output voltage of around 3.3V compared to 3.5V for the two inputs. The reason for this is due to the reference voltage for the opamps changing slightly depending on the input voltage which is covered in the next section.

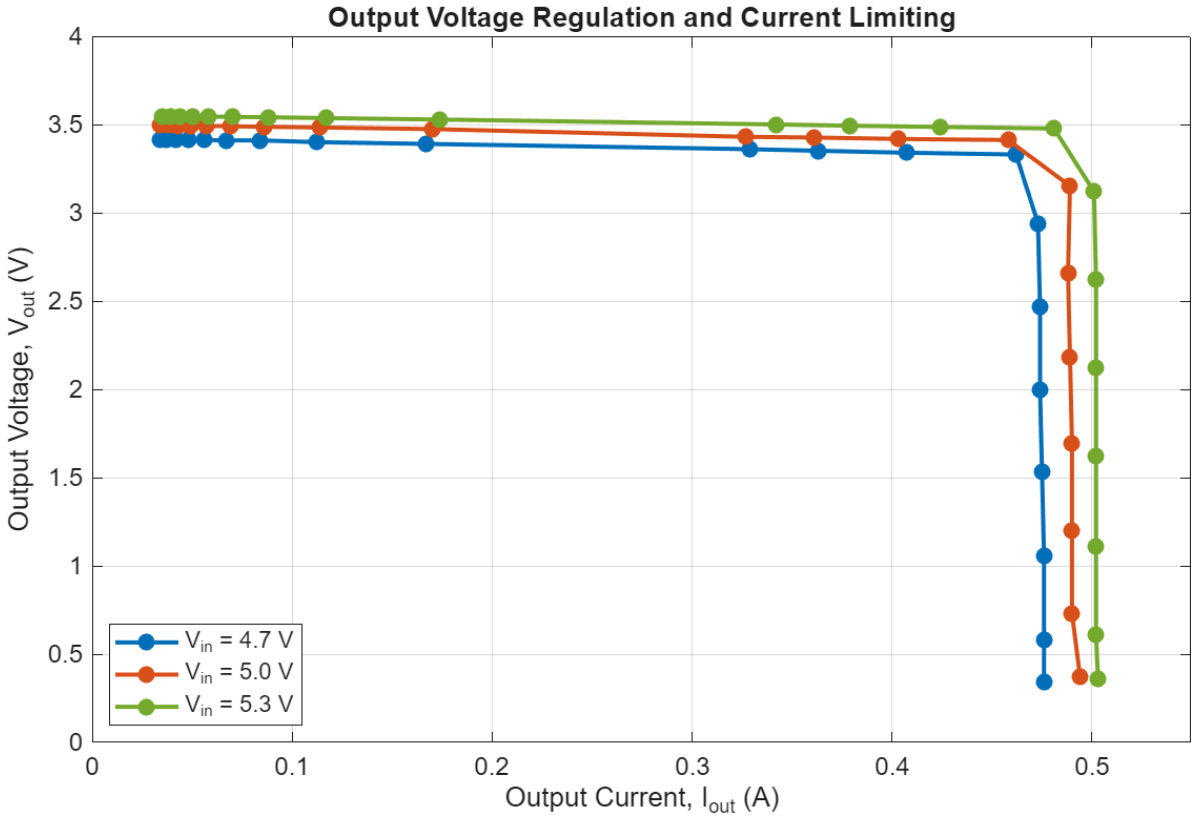


Figure 1: 0.5A current limiting at different input load voltages

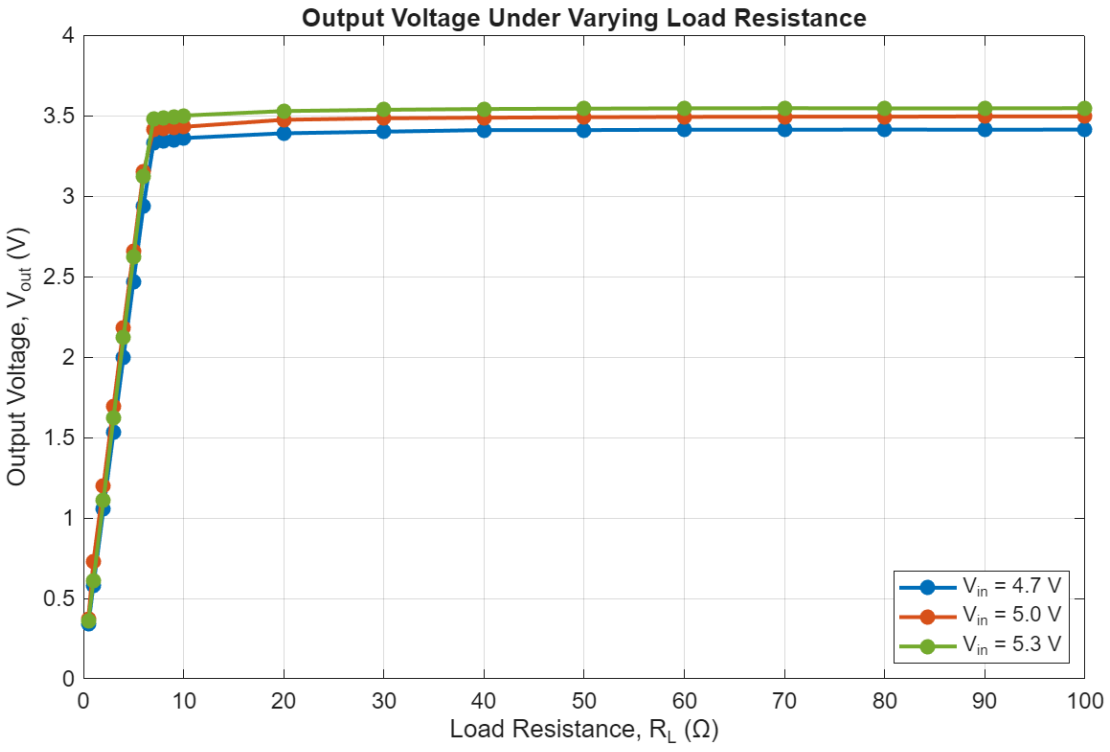


Figure 2: Output deviation from 3.5V due to different input load voltages

Vref protection:

Our reference voltage for our opamps were tested under different input voltages from the power supply in order to see if a sudden increase in voltage would cause our reference voltage to increase dramatically and damage the opamp. After testing with a few voltages below and above the expected 5V input we can see that the reference voltage did vary up and down but remained relatively close to 1.75V. The change in voltage is due to the zener diode not clamping perfectly. Different input voltages drive different amounts of current into the zener diode causing its voltage to vary slightly as shown in the table below.

Table 5: Opamp reference voltage due to different input voltages

Input Voltage	Vref
4.7V	1.717V
5V	1.752V
5.3V	1.783V
7V	1.9V

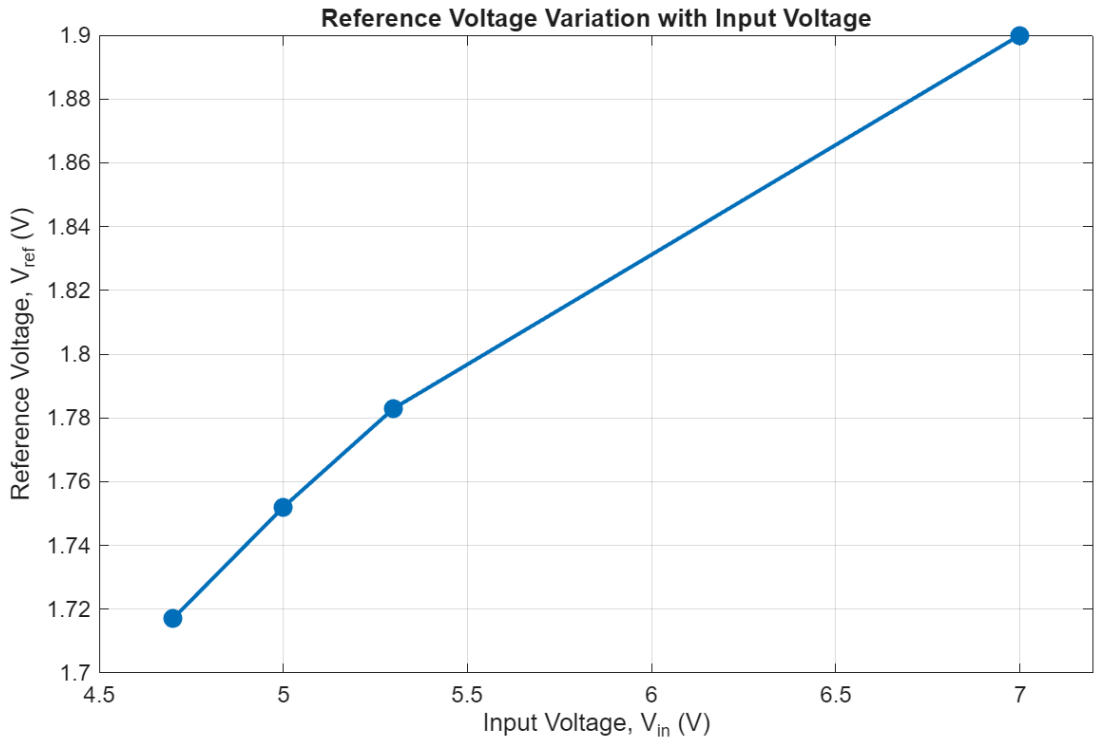


Figure 3: Vref voltage variation with different input voltages

Voltage Ripple:

Our goal was to achieve a voltage ripple within 5% of our output voltage. Our output voltage was approximately 3.5V leaving us with a target peak to peak output ripple of around 175mV or below. After testing on the oscilloscope we found the average ripple to be around 140mV. The ripple of output also causes us to adjust our overvoltage reference to a slightly higher voltage in order to stop top false triggering due to noise or small sharp spikes. When testing the out output voltage we noticed the breaker was sometimes disconnecting the circuit which led us to increase the trip voltage for our breaker.

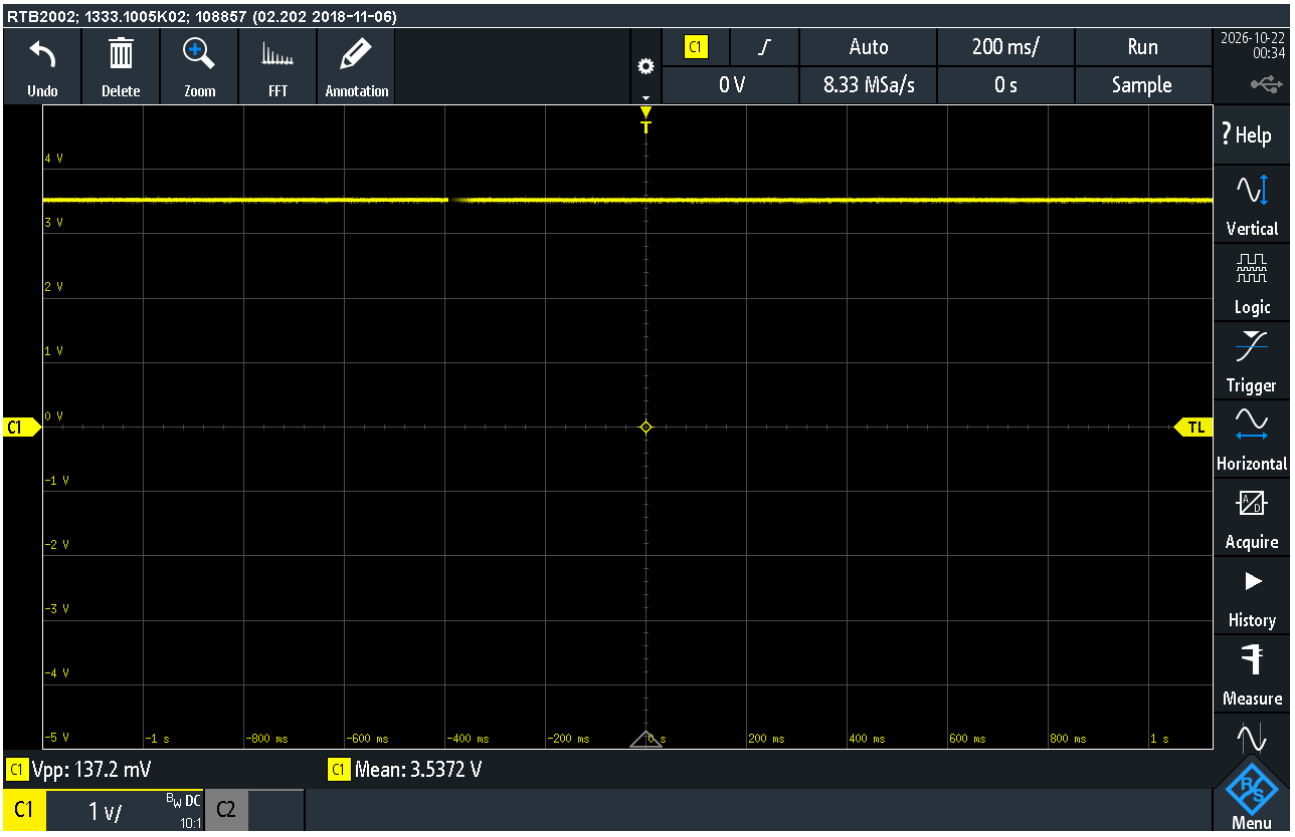


Figure 4: Output voltage ripple measurement

Mosfet Temperature:

The mosfet we used was the IRF4905 which has a Junction to ambient rating of 62°C/W and a max operating temperature of 175°C. When looking at the worst scenario for our circuit at a 5V input and 0.36V output we achieve a theoretical temperature of 166°C disregarding the heatsink we have on. Even at the worst case scenario we are still below the operating limit and during normal operation and a steady 3.5V output as well as the effect of the heatsink will keep the mosfet temperature comfortably before the limit.

Overvoltage Protection:

When testing the overvoltage protection we noticed that the breaker would disconnect the circuit at random times. After testing the voltage it was receiving we noticed that the oscilloscope showed small spikes in the output voltage causing the breaker to false trigger. To counteract this problem we decided to increase the threshold voltage the comparator trips at to 3.7V in order to leave it isolated from a regular output and its voltage ripple. When testing the functionality of the breaker we connected a 3.7V input on a second channel of the power supply and fed it to the output voltage to see how it behaved. Shown below is an oscilloscope waveform showing us tripping the breaker and re-enabling it using a push button to show its functionality.

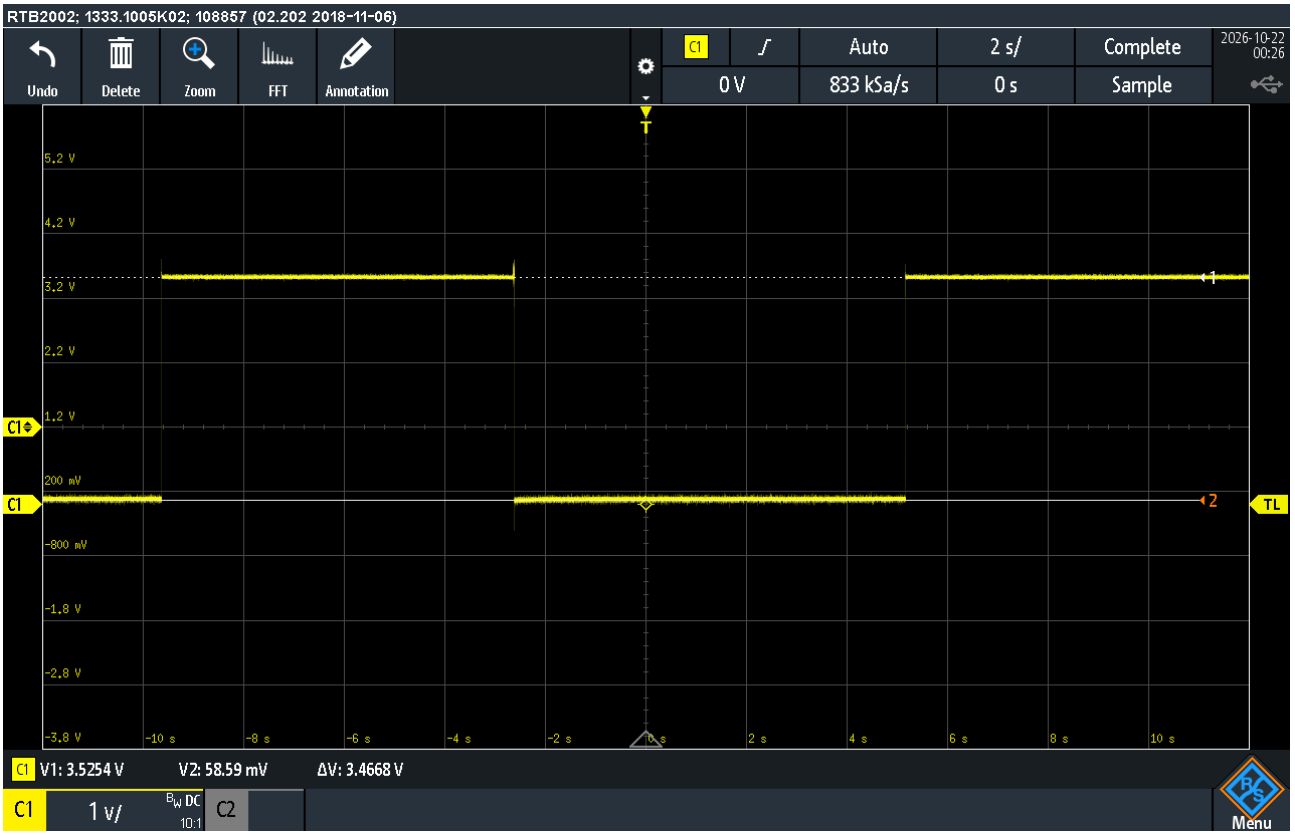
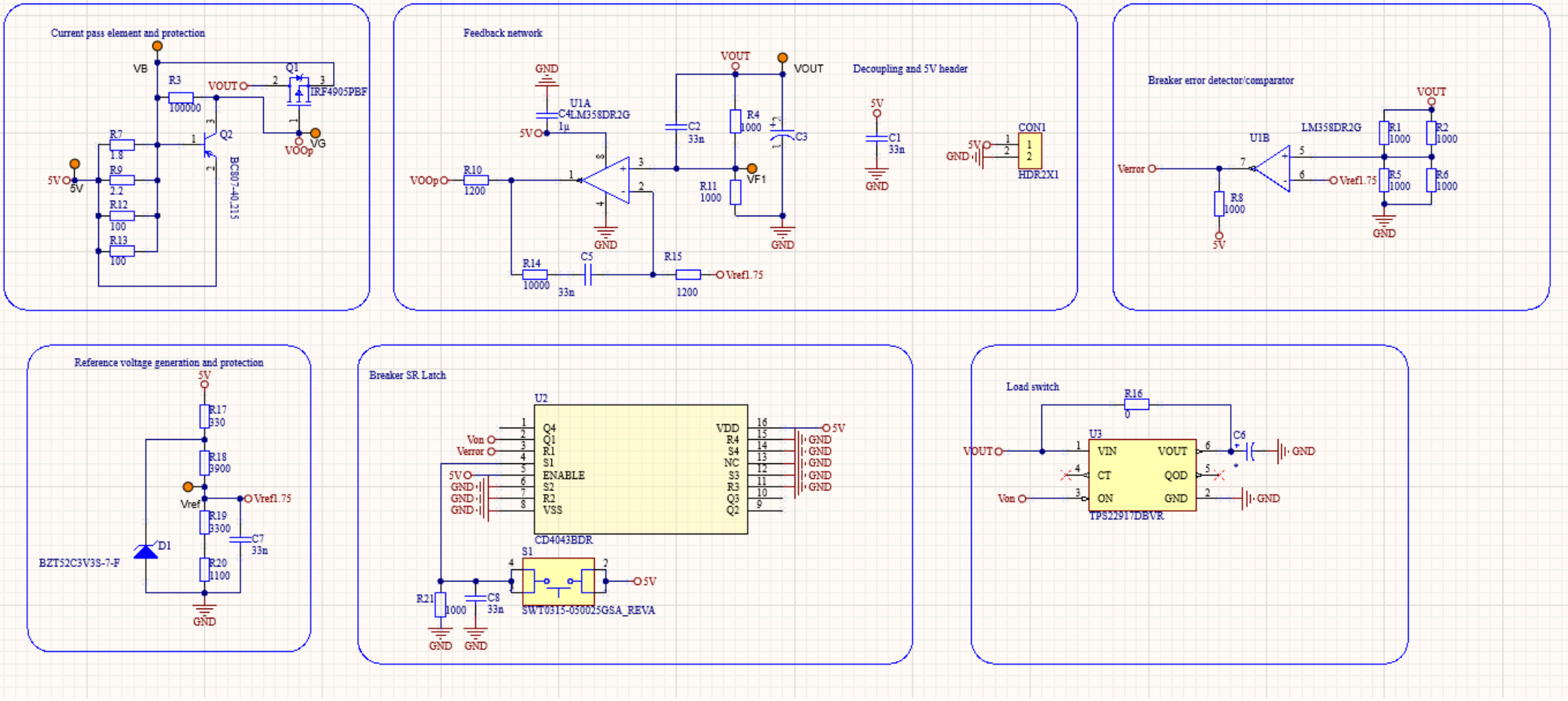
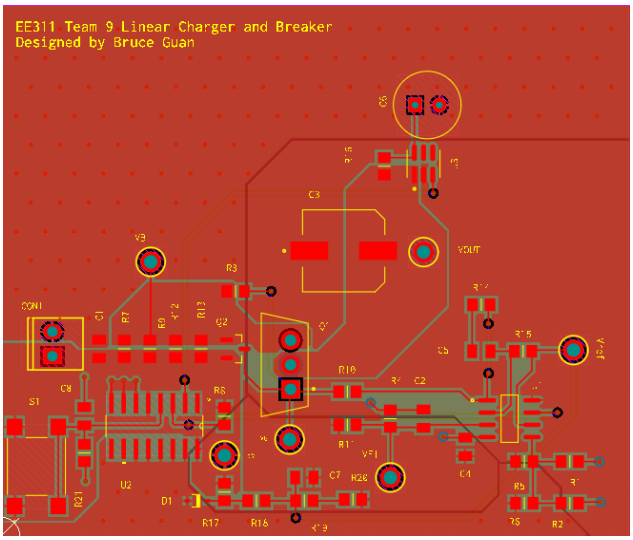
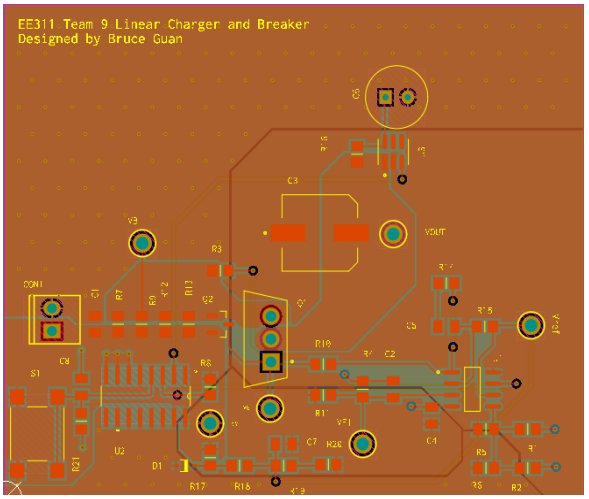


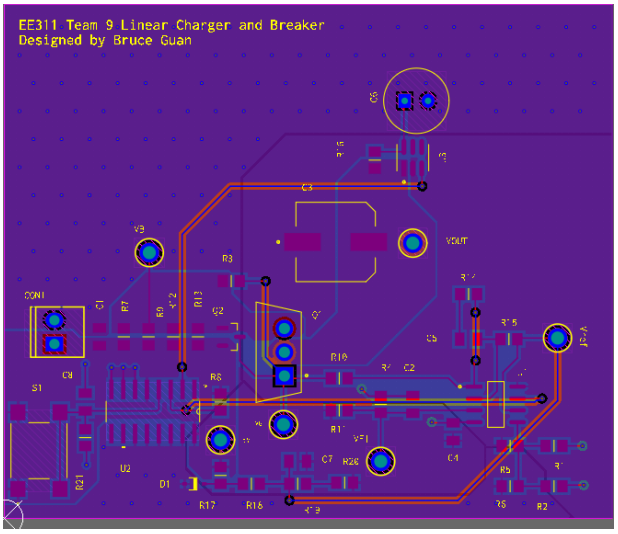
Figure 5: Breaker disconnecting output voltage and push button re-enabling breaker

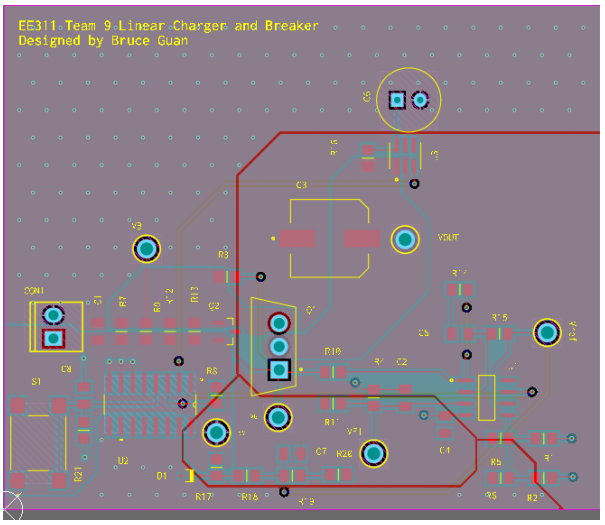
ELECTENG 311 Team 9 Linear Charger and Breaker Schematic




EE311 Team 9 Linear Charger and Breaker
Designed by Bruce Guan


EE311 Team 9 Linear Charger and Breaker
Designed by Bruce Guan


EE311 Team 9 Linear Charger and Breaker
Designed by Bruce Guan


EE311 Team 9 Linear Charger and Breaker
Designed by Bruce Guan

PCB Layout and Layer Design

Layout: Components were arranged according to their functional blocks to minimise trace lengths and simplify routing. The bulk output capacitor, circuit breaker switch, supercapacitor and charger output were deliberately placed close together to share a Vout copper pour with the MOSFET drain (series pass element). This reduces trace resistance and voltage drop in the high-current output path.

Layers: A four layer PCB was selected instead of a conventional two layer design, with the following configuration:

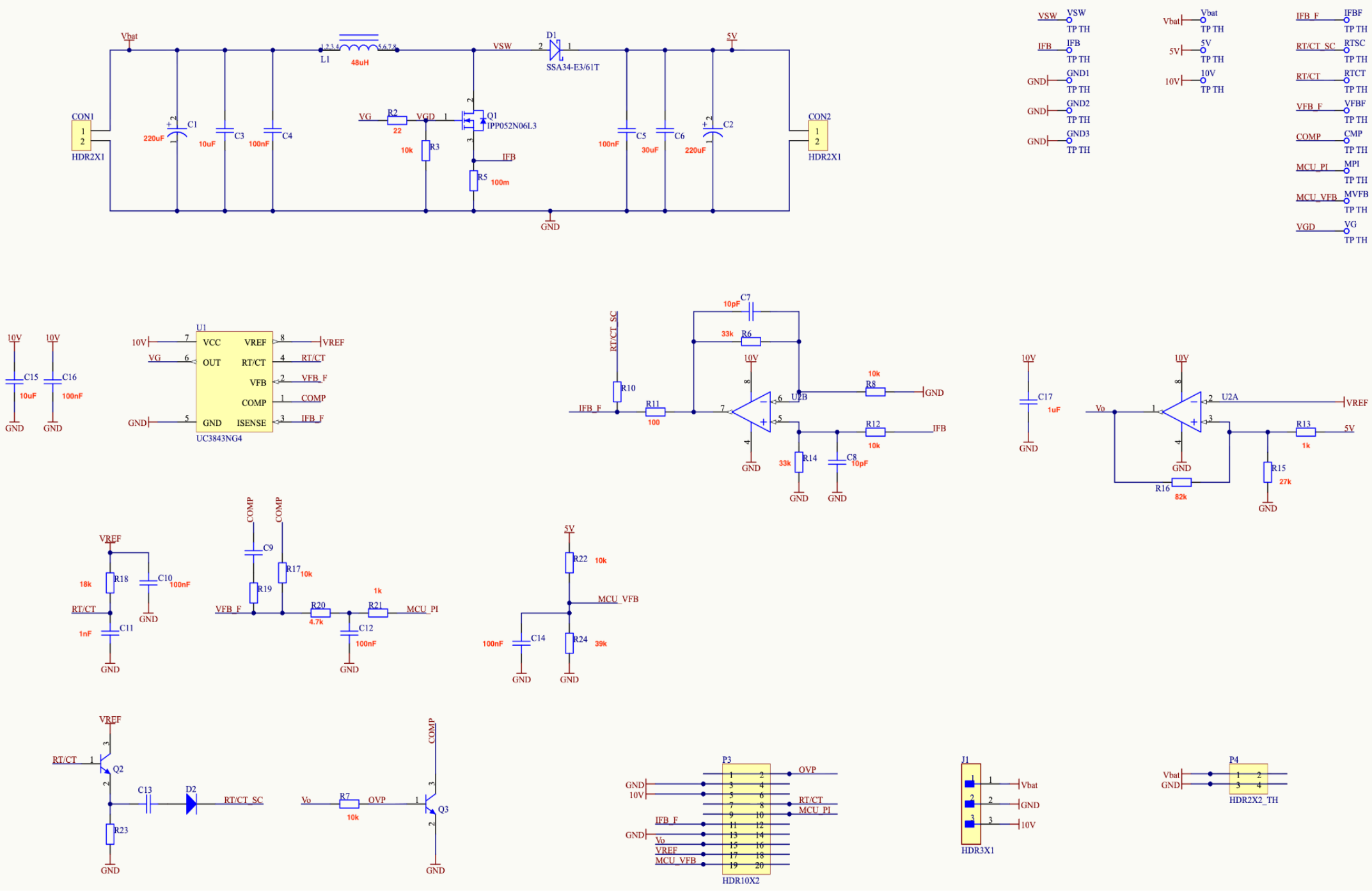
Top Layer: Contains most signal traces and component connections, minimising the need for vias when routing SMD components. Dedicated 5 V and Vout copper pours were also implemented to reduce trace resistance, parasitic inductance and voltage drop, particularly in high current paths.

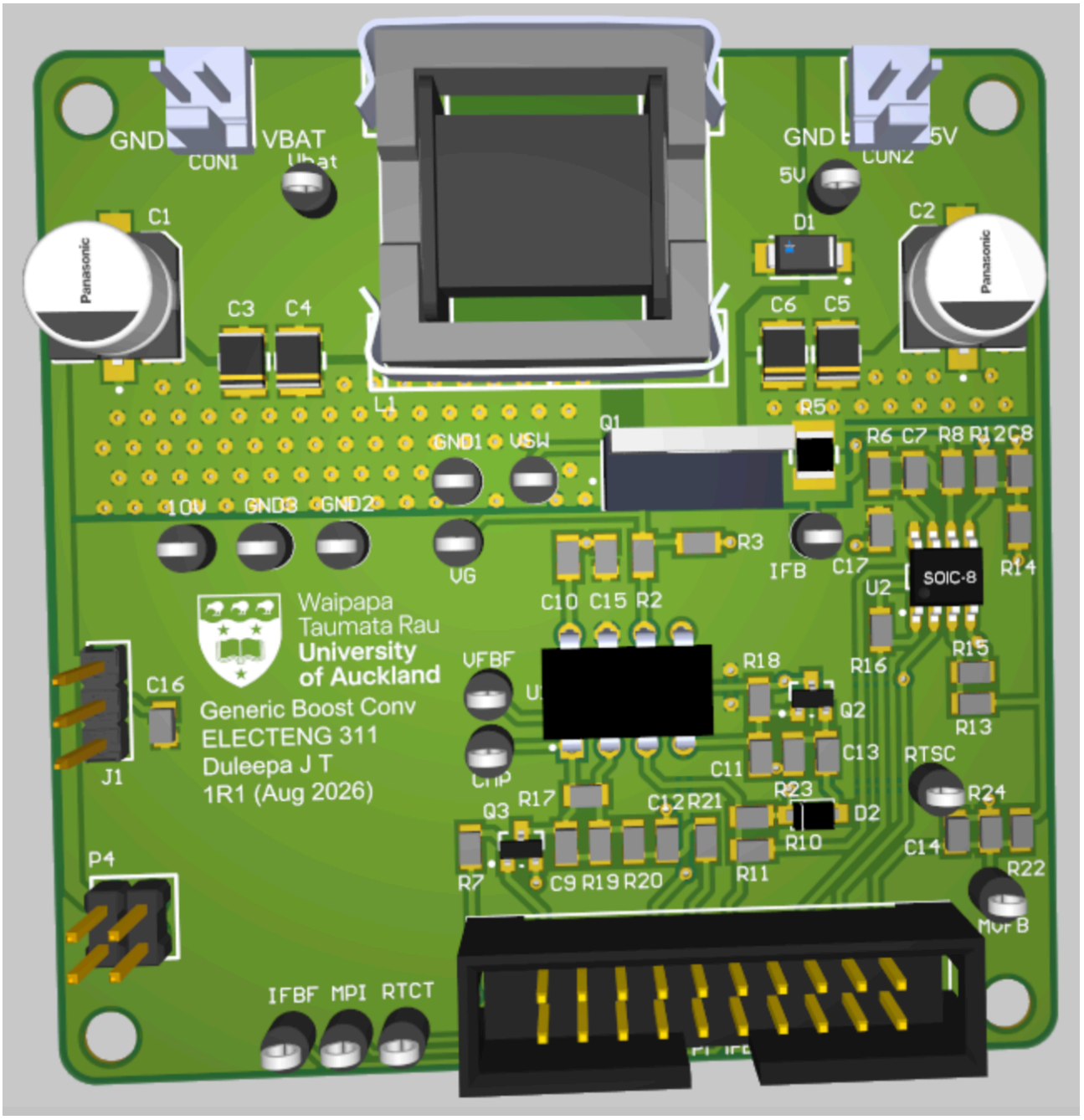
Layer 2 (GND Plane): An uninterrupted GND plane provides a low impedance return path for signals on the top layer, minimising current loop area and reducing electromagnetic interference. It also provides separation between the top signal layer and the third power distribution layer, reducing noise coupling.

Layer 3 (Power Plane): Contains large 5 V and Vout copper pours for power distribution. These are connected to corresponding pours on the top layer using vias placed close to components, allowing power connections where direct top layer routing is restricted. The larger copper areas help reduce resistance and parasitic inductance.

Bottom Layer: Reserved for signal traces that would otherwise require longer or more complicated routing on the top layer. Routing is kept minimal to preserve GND copper coverage and provide additional return paths through connections to Layer 2.

Ground Pour and Via Stitching: GND was selected as the default copper pour on all four layers wherever space was available. This provides additional low impedance return paths, simplifies ground connections through thermal reliefs and reduces routing complexity. Ground stitching vias were placed throughout suitable GND regions to electrically connect the ground copper across layers, reduce return path impedance and assist with heat spreading through the PCB.


Vbat
L1
48uH
VSW
D1
SSA34-E3/61T
5V
CON1
HDR2X1
C1
220uF
C3
10uF
C4
100nF
VG
R2
22
VGD
Q1
1PP052N06L3
R3
10k
IFB
R5
100m
C5
100nF
C6
30uF
C2
220uF
CON2
HDR2X1
GND
VSW
IFB
GND1
GND2
GND3
TP TH
Vbat
5V
10V
IFB_F
RT/CT_SC
RT/CT
VFB_F
COMP
MCU_PI
MCU_VFB
VG
IFBF
RTSC
RTCT
VFBF
CMP
MPI
MVFB
C15
10uF
C16
100nF
U1
VCC
OUT
GND
VREF
RT/CT
VFB
COMP
ISENSE
UC3843NG4
C7
10pF
R6
33k
R10
R11
100
U2B
R8
10k
R12
10k
R14
33k
C8
10pF
C17
1uF
U2A
R13
1k
R15
27k
R16
82k
Vo
R18
18k
C10
100nF
C11
1nF
C9
R17
10k
R19
R20
4.7k
R21
1k
C12
100nF
R22
10k
C14
100nF
R24
39k
Q2
C13
D2
R23
R7
10k
OVP
Q3
P3
HDR10X2
J1
HDR3X1
P4
HDR2X2_TH


GND
VBAT
CON1
Vbat
GND
5V
CON2
5V
C1
C2
D1
C3
C4
C6
C5
L1
R5
Q1
GND1
VSW
R6
C7
R8
R12
C8
10V
GND3
GND2
VG
R3
IFB
C17
SOIC-8
R14
U2
C10
C15
R2
R18
R16
R15
VFBF
Q2
C16
R13
C13
C11
RTSC
R23
R24
R17
Q3
C12
R21
D2
R10
C14
R22
R7
C9
R19
R20
R11
MVr B
J1
P4
IFBF
MPI
RTCT
Waipapa
Taumata Rau
University
of Auckland
Generic Boost Conv
ELECTENG 311
Duleepa J T
1R1 (Aug 2026)

Boost Converter

Team Members: Alexander Ward - Everything

Order of testing and reasoning.

1. OVP was tested first because it protects the converter and load during later tests.
2. Current sensing was checked with the UC3843 removed and a generator driving the gate, preventing a sensing fault from causing runaway or damage.
3. Open-loop tests measured duty, losses and ripple across loads without feedback, establishing a baseline for closed-loop operation.
4. Closed-loop testing added the PI firmware last, isolating controller/interface issues from known hardware behaviour.

Table 1: Test conditions.

Stage	Level	Hardware under test	Firmware	Supply and load	Instruments
1. OVP	Block	UC3843 fitted; OVP comparator and COMP node live	None in the loop	12 V rail; bench supply on the 5 V test point in place of Vout, swept through the threshold in both directions	RTB2002 Scope
2. Current measurement	Block	Op-amp stage and filter; UC3843 removed	None in the loop	12 V rail for the op-amp only; Vbat 3.5 V, 16.25 Ω load; generator on Vg: 5 Vpp, 2.5 V offset, 45 % duty, 100 kHz	RTB2002 Scope, DMM
3. Open loop	Combined blocks	Power stage, UC3843 and OVP	None; duty set by the signal generator on RT/CT	Vin 2.5 V and 3.5 V; 5, 16.25, 27.5, 38.75 and 50 Ω; 12 V rail	RTB2002 Scope, DMM, supply and electronic load readings
4. Closed loop	Final subsystem	Full boost converter with current sense and OVP active	PI controller implemented using ATmega328PB with ADC reference of 5.04V	Vbat 2.5 V and 3.5 V; 4.75–52.5 Ω; 12 V rail	RTB2002 Scope, DMM, supply and load displays

1. Over-Voltage Protection (OVP)

1.1 Test plan

Aim: verify the OVP trip level, the release level (hysteresis) and the response speed of the comparator acting on the COMP node.

- UC3843 fitted, 12 V rail powered, so the protection circuit and COMP node are live.
- A bench DC supply drives the 5 V test point in place of Vout. It was swept slowly through the threshold in both directions.
- Oscilloscope (R&S RTB2002): C1 probes the Vout test point, C2 probes COMP. Single-shot captures were triggered on the COMP edge, with cursors on the C1 level at the start of the COMP transition.
- Trip: C1 raised slowly through about 5.25 V. Release: C1 lowered slowly through about 5.1 V. Each direction was captured several times to check repeatability.
- Speed (1): measured as the time between the COMP (C2) pin beginning its drop to the time it crosses the zero axis
- Speed (2): use a pulsed input to measure the time taken for the whole system to react from OVP crossing to the time the COMP pin drops to zero.

1.2 Results

When C1 exceeds the upper threshold, COMP falls from about 0.8 V to ground and remains low until C1 falls below the release threshold. It then returns to about 0.7–0.9 V, confirming hysteresis.

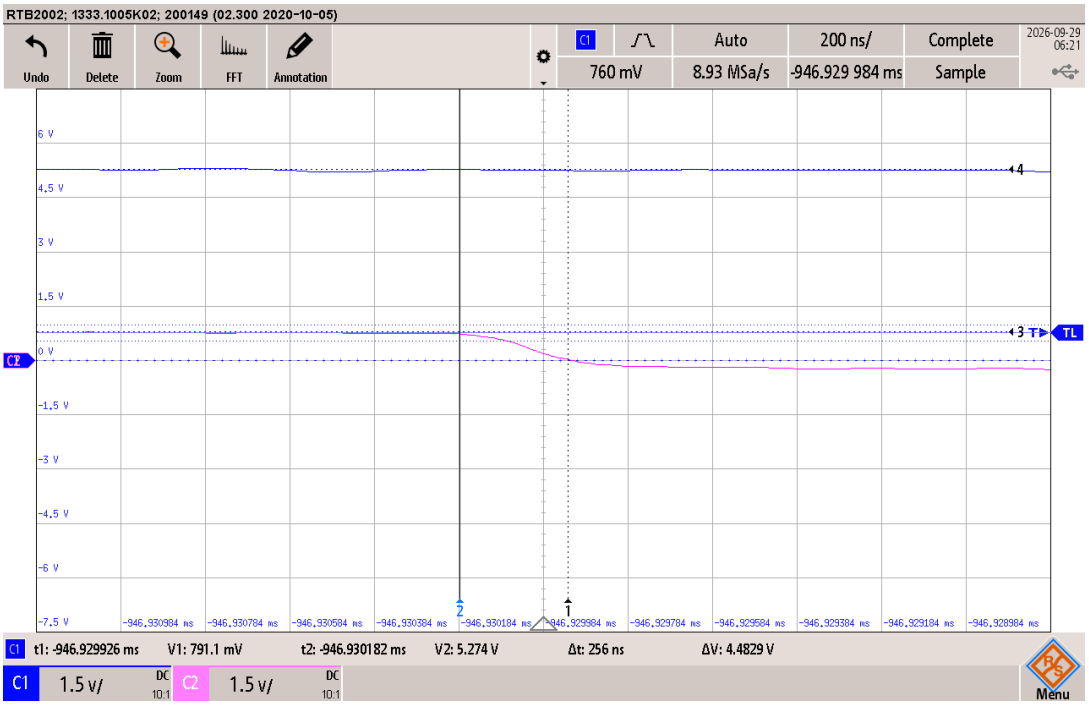


Figure 1: OVP trip. C2 (COMP) falls as C1 reaches 5.274 V. Cursor Δt = 256 ns is the time taken for the COMP pin to reach zero.

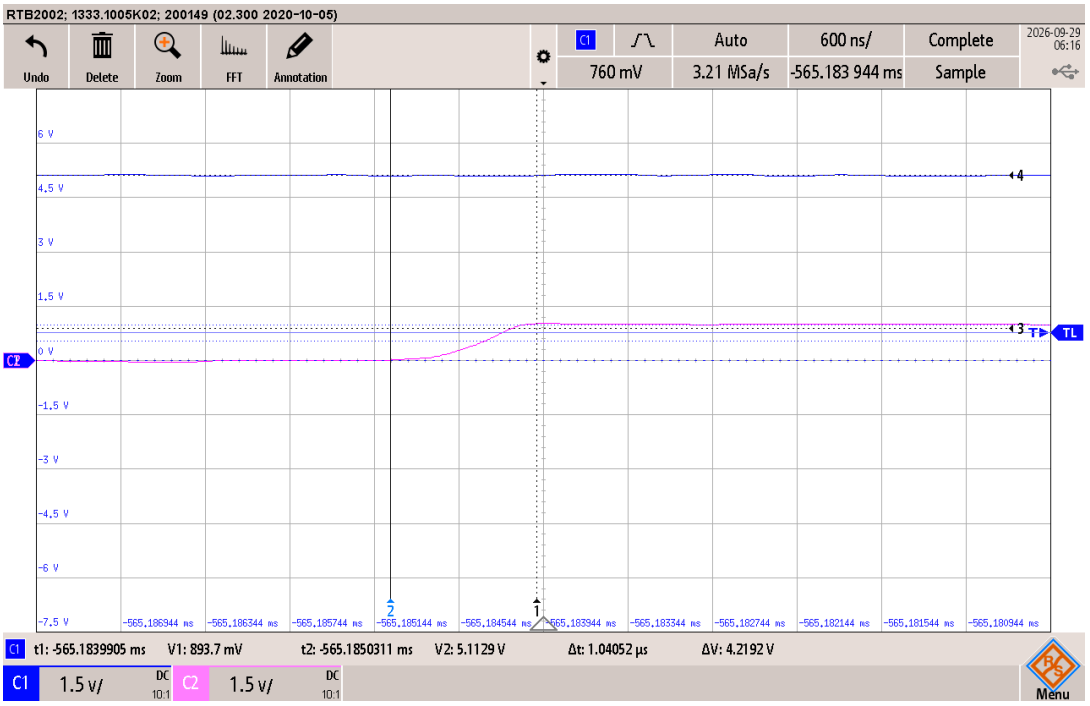


Figure 2: OVP release. C2 (COMP) rises as C1 falls to 5.113 V. Cursor Δt = 1.04 µs is the time taken for the COMP pin to climb back to its stable operational voltage.

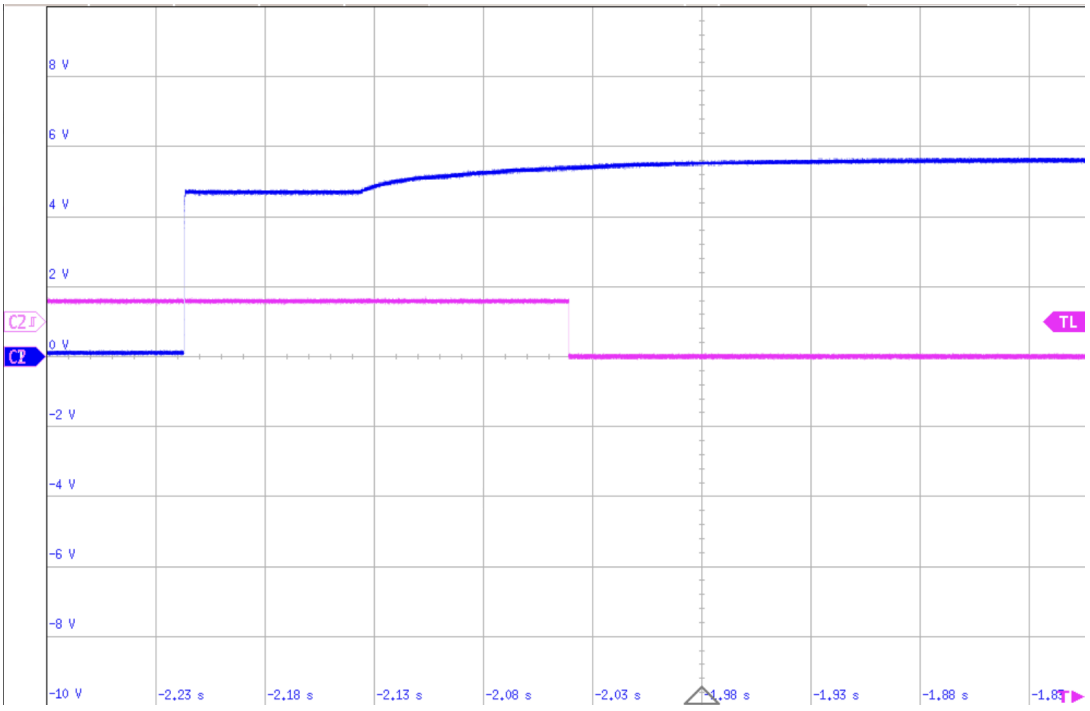


Figure 3: OVP reaction to pulse. Input pulse edge: approximately 2.217 s, OVP reacts at approximately 2.04112 s, meaning it reacts to a voltage spike in approximately 176 ms.

Table 2: OVP test results

Quantity	Oscilloscope	Theory	Deviation from Theory
Trip level (C1 rising)	5.274 V	5.246 V	+0.54%
Release level (C1 falling)	5.113 V	5.1 V	+0.25%
Hysteresis	0.161 V	0.146 V	+10.2%
COMP pin, fall time	256 ns		
COMP pin, rise time	1.04 µs		
Input pulse reaction time	176 ms		

2. Current Measurement (OpAmp and Filter)

2.1 Test plan

Aim: characterise the DC gain and the dynamic response of the OpAmp stage and its output filter.

- UC3843 is removed and 12 V rail supplied for the OpAmp only, so the feedback control does not interfere with the clean oscilloscope generated signal that is driven directly to Vg.
- An arbitrary load and signal are applied to the boost converter, in this case,
    - Vbat = 3.5 V, load 16.25 Ω.
    - Function generator: 5 Vpp, 2.5 V offset, 45 % duty, fs = 100 kHz.
- Probes data from IFB (sense signal before the amplifier) and IFBF (amplified and filtered output) are then plotted to demonstrate the amplification and response.

2.2 Results

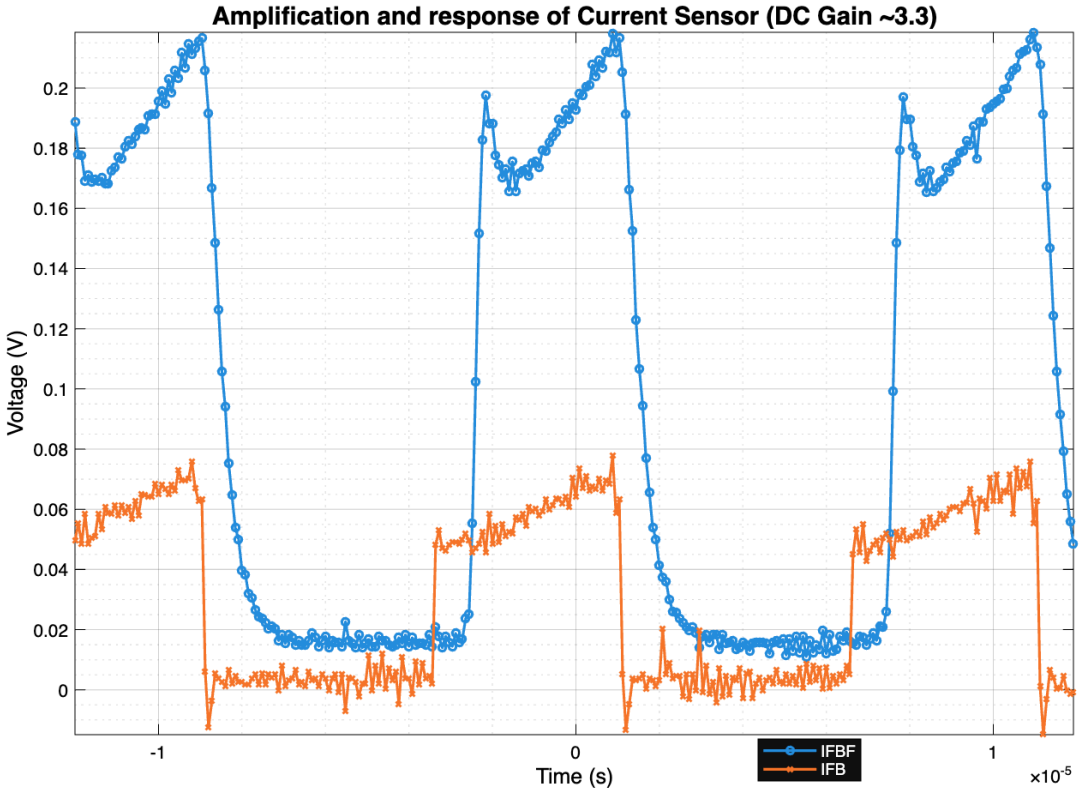


Figure 4: IFB and IFBF over two 10 µs periods. IFB is the raw sense signal and IFBF is the amplified, filtered output.

2.3 Discussion

Gain. The measured DC gain was 3.28, close to the design value of 3.3. Waveform measurements indicate a gain of about 3 at 100 kHz. The filter smooths switching noise but introduces delay: IFBF follows the rising edge by about 1.07 µs and the falling edge by about 0.24 µs. This delay is small relative to the 10 µs switching period. This is a reasonable trade-off relative to the 10 µs switching period.

3. Open Loop Various Load Testing

3.1 Test plan

Aim: measure the open-loop duty cycle needed to hold a 5 V output, and the resulting efficiency, across a range of loads at both ends of the battery range, with no form of feedback control.

- Vin = 2.5 V and 3.5 V; load resistance 5, 16.25, 27.5, 38.75 and 50 Ω.
- UC3843 fitted and powered from 12 V rail while VFB and ISENSE are pinned to ground so as to not interfere with the signal generated duty cycle being driven directly into RT/CT.
- At each point the duty cycle was adjusted until Vout (probed with an oscilloscope on the 5 V (Vout) test point) was 5 V. Pin and Pout were recorded from the DC power supply and electronic load to determine efficiency = Pout / Pin.
- The duty cycle is set on the signal generator, which is inverted relative to the real duty cycle, so the real duty is 100% minus the generator setting. The real duty is plotted.

3.2 Results

Table 3: Open-loop results

Vin (V)	Vout (V)	Load (Ω)	SG duty (%)	Real duty (%)	Pin (W)	Pout (W)	Eff. (%)
2.5	4.7*	5	25	75	6.95	3.3	47
2.5	5	16.25	41	59	1.71	1.35	79
2.5	5	27.5	43	57	1.00	0.83	83
2.5	5	38.75	44	56	0.70	0.60	86
2.5	5	50	44	56	0.55	0.48**	87
3.5	5	5	55	45	5.16	3.4	66
3.5	5	16.25	61	39	1.61	1.32	82
3.5	5	27.5	63	37	0.97	0.84	87
3.5	5	38.75	64	36	0.71	0.62	87
3.5	5	50	65	35	0.55	0.49**	89

* At 2.5 V and 5 Ω the converter failed to reach 5 V; this point was taken at approximately 4.75 V.

** At the lightest load (50 Ω) the circuit delivers 0.48 W at 2.5 V and 0.49 W at 3.5 V, which is 4 % and 2 % below the 0.5 W minimum.

3.3 Discussion

The required duty cycle decreased as load resistance increased and was higher at 2.5 V than at 3.5 V. At light load, measured duty (56 % and 35 %) exceeded ideal values (50 % and 30 %) because of converter losses. Efficiency increased with load resistance, reaching 87 % at 2.5 V and 89 % at 3.5 V for 50 Ω. The 3.5 V input was more efficient at every load. At 2.5 V and 5 Ω, the output reached only about 4.75 V despite 75 % duty, indicating that the converter could not supply the required power at this operating point. At 50 Ω, output power was 2–4 % below the 0.5 W minimum.

4. Closed Loop Testing

4.1 Test plan

Aim: verify 5 V regulation, efficiency and output ripple with the ATmega328PB PI controller over 2.5–3.5 V input and 4.75–52.5 Ω load (about 0.5–5 W).

- Controller: the PI (Kp = 0.5, Ki = 200, 1 kHz sample rate) reads Vout through a 10 kΩ / 39 kΩ divider (ratio 0.796, ADC reference 5.04 V). Its output is a 15.625 kHz PWM on PB1 (PWM PI), limited to 1.45–3.5 V (28.8–69.4 % duty) and filtered by R21/C12 before it reaches the UC3843 feedback input.
- Unlike the open-loop tests in Section 3, VFB and ISENSE are not tied to ground and the PI sets the duty. The UC3843 is powered from the 12 V rail.
- The supply was adjusted so that Vbat at the converter's own terminals was 2.5 V or 3.5 V. The supply setting was also recorded, because wiring loss reached 330 mV at 2.5 V / 4.75 Ω.
- The electronic load was in constant-resistance mode at 4.75, 5, 16.25, 27.5, 38.75, 50 and 52.5 Ω.
- Instruments: DMM for Vbat, Vout (at the output capacitor), the PWM PI mean and MCU_VFB; the supply display for Iin; the electronic load display for Iout; the oscilloscope for ripple, PWM PI duty and PWM PI frequency.
- Pin = Vbat × Iin, Pout = Vout × Iout, and efficiency = Pout / Pin. An efficiency that includes wiring loss, using the supply setting instead of Vbat, is also given.
- Ripple was measured AC coupled, with a 20 MHz bandwidth limit at the output capacitor. It is reported two ways: with cursors on the ripple body, excluding switching spikes (the value used for comparison), and with the oscilloscope's automatic peak-to-peak measurement, which includes the spikes.

4.2 Results

Table 4: Closed-loop results at Vin = 2.5 V

R (Ω)	Vout (V)	Iout (A)	Pout (W)	η (%)	η incl. leads (%)	Ripple, cursors (mV)	Ripple, scope (mV)	PWM PI duty (%)
4.75	4.970	1.010	5.020	78.3	69.2	285	540	29.4
5.00	5.010	0.953	4.775	76.8	69.8	285	500	32.0
16.25	5.020	0.303	1.521	85.7	83.6	114	377	52.3
27.5	5.021	0.180	0.904	86.1	84.7	79	482	56.4
38.75	5.020	0.128	0.643	86.9	86.1	72	291	58.5
50.0	5.020	0.099	0.497	87.2	86.9	61	317	59.6
52.5	5.020	0.095	0.477	88.0	87.2	69	405	59.8

Table 5: Closed-loop results at Vin = 3.5 V

R (Ω)	Vout (V)	Iout (A)	Pout (W)	η (%)	η incl. leads (%)	Ripple, cursors (mV)	Ripple, scope (mV)	PWM PI duty (%)
4.75	5.010	0.991	4.965	86.3	82.7	228	428	44.6
5.00	5.010	0.944	4.729	87.1	83.5	200	640	45.7
16.25	5.020	0.303	1.521	89.0	87.8	101	550	57.5
27.5	5.020	0.180	0.904	89.3	88.8	79	560	60.1
38.75	5.029	0.128	0.644	89.2	88.8	79	519	61.7
50.0	5.020	0.100	0.502	89.7	89.4	58	370	62.9
52.5	5.018	0.096	0.482	91.8	91.5	58	370	63.0

Table 6: Summary of closed-loop results

Quantity	Vin = 2.5 V	Vin = 3.5 V
Vout range	4.970–5.021 V	5.010–5.029 V
Largest Vout error	−0.6 %	+0.6 %
Load regulation (52.5 Ω to 4.75 Ω)	50 mV (1.0 %)	8 mV (0.16 %)
Efficiency range	76.8–88.0 %	86.3–91.8 %
Average efficiency	84.1 %	88.9 %
Largest ripple (cursors)	285 mV	228 mV
MCU_VFB range	3.994–4.000 V	3.988–4.000 V
PWM PI duty range	29.4–59.8 %	44.6–63.0 %

Line regulation was 9 mV or less at each load except 4.75 Ω, where Vout differed by 40 mV between input conditions. At 50 Ω, output power was 0.497 W at 2.5 V and 0.502 W at 3.5 V.

The PB1 PWM output was measured on the oscilloscope at 15.62 kHz, swinging from 0 to about 5 V. Figure 5 shows the waveform at 62.8 % duty with a 52.5 Ω load.

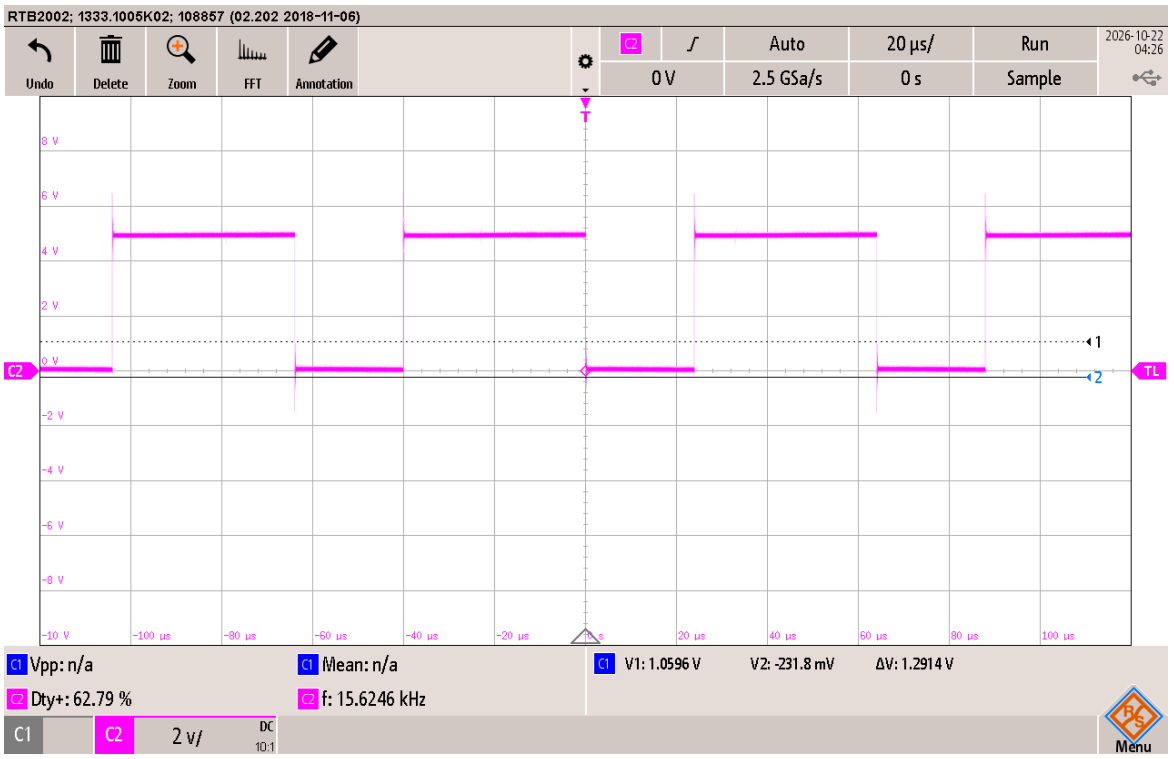


Figure 5: PWM PI (PB1) PWM output, 0–5 V, 15.62 kHz, 62.8 % duty, 52.5 Ω load. 2 V/div, 20 µs/div.

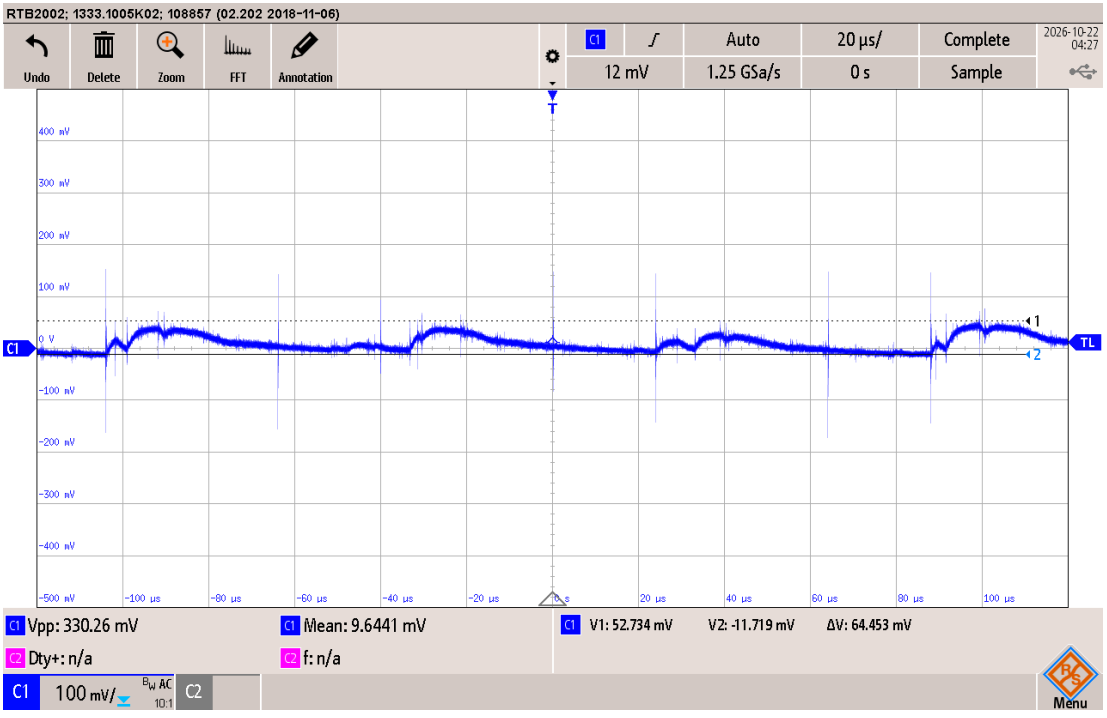


Figure 6: Output ripple at 52.5 Ω, AC coupled. Automatic Vpp = 330 mV (includes spikes); cursor ΔV = 64.5 mV (ripple body). 100 mV/div, 20 µs/div.

Table 7: Open-loop and closed-loop efficiency

R (Ω)	2.5 V open loop (%)	2.5 V closed loop (%)	3.5 V open loop (%)	3.5 V closed loop (%)
5	47	76.8	66	87.1
16.25	79	85.7	82	89.0
27.5	83	86.1	87	89.3
38.75	86	86.9	87	89.2
50	87	87.2	89	89.7

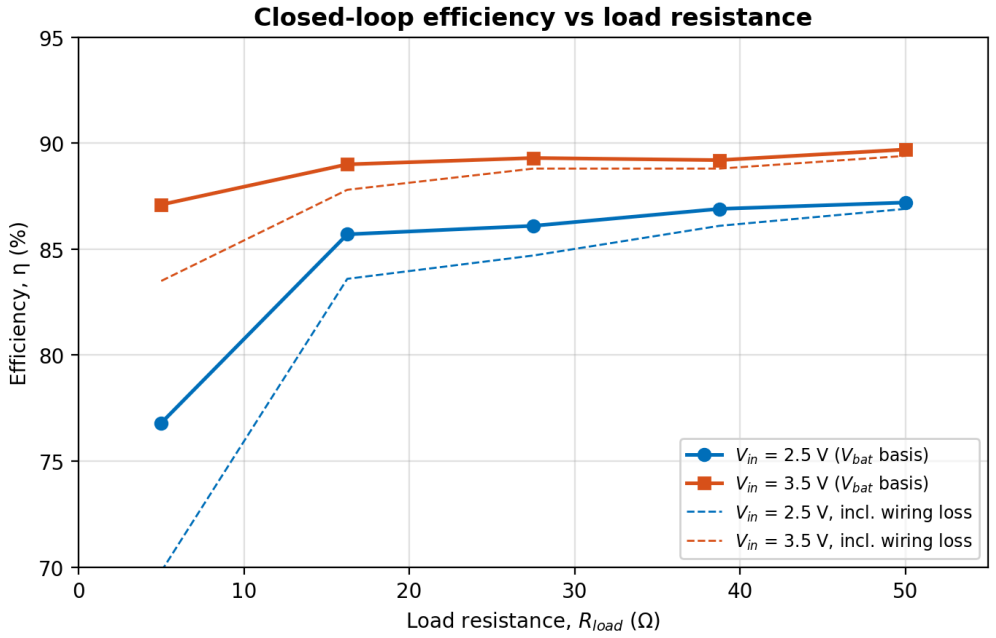


Figure 7: Closed-loop efficiency vs load resistance. Solid lines use Vbat at the converter terminals; dashed lines include wiring loss (supply setting).

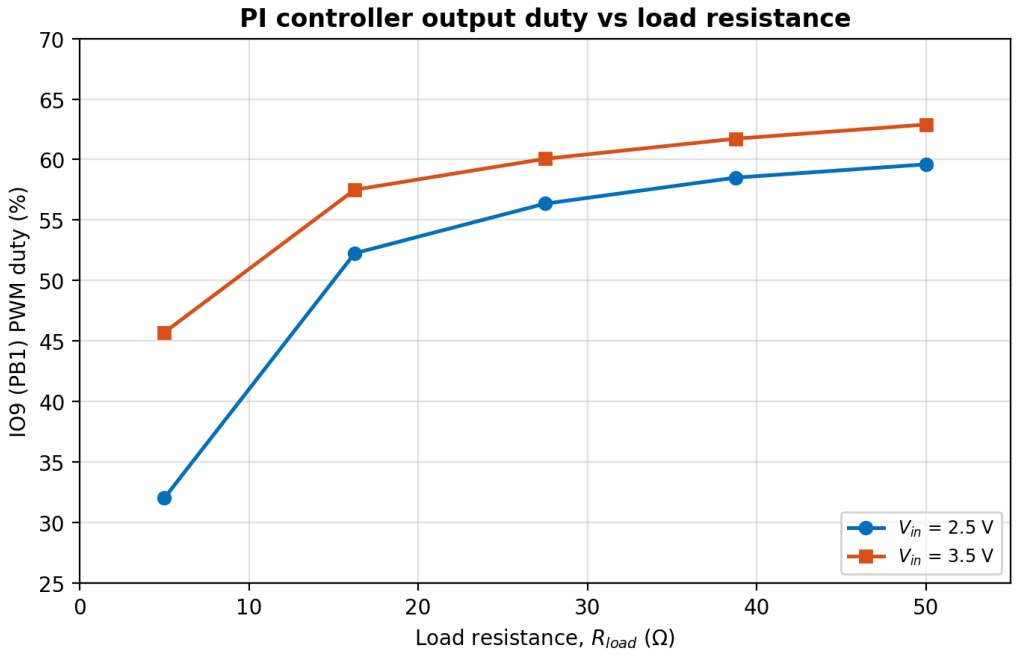


Figure 8: PI controller output (PB1 PWM) duty vs load resistance.

4.3 Discussion

Regulation. Across 14 operating points, Vout was 4.970–5.029 V (within ±0.6 % of 5 V). Load regulation was 50 mV at 2.5 V input and 8 mV at 3.5 V. Voltage drop between the output capacitor and feedback divider was most evident at 2.5 V / 4.75 Ω.

Controller headroom. PWM PI duty remained within its limits, but reached 29.4 % at 2.5 V / 4.75 Ω, close to the 28.8 % lower limit. This is the limiting operating point.

Efficiency. Closed-loop efficiency was 76.8–88.0 % at 2.5 V and 86.3–91.8 % at 3.5 V. Including wiring losses reduced efficiency at heavy 2.5 V loads to about 69–70 %. Open- and closed-loop results agreed within about 2.5 percentage points from 38.75 Ω upwards.

Light-load switching. At 3.5 V / 52.5 Ω, switching was 100 kHz but gate pulses were irregular. Coupling from the 15.62 kHz MCU PWM through R21/C12 is a possible cause, but remains unconfirmed.

Limitations. Thermal performance of MOSFET and inductor temperatures are yet to be confirmed.

Table 8: Proposed specifications and measured performance

Parameter	Specification	Measured	Result
Input voltage range	2.5–3.5 V	Operated across 2.5–3.5 V	Met
Output voltage	5 V	4.970 V (2.5 V, 4.75 Ω) to 5.018 V (3.5 V, 52.5 Ω)	Met (+/- 0.6%)
Output power	0.5–5 W	0.497 W (2.5 V) at 50 Ω; up to 5.02 W at 4.75 Ω	Met, except 0.6 % below 0.5 W at 2.5 V
Output ripple	50mV max	58–285 mVpk-pk (cursors); 291–640 mVpk-pk (scope automatic)	Not met (lowest 58 mV)
Load Tolerance	5%	4.97V for 2.5V at 4.75 Ω and 5.018V for 3.5V at 52.5 Ω	Met